

Matthew Perryman

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Profile

Engineer focused on dynamical systems under uncertainty. Background across electrical and computer systems engineering, econometrics, control systems, optimisation, and high-dimensional industrial time-series data. Interested in modelling, diagnosing, and improving complex systems where noisy data, constraints, feedback, uncertainty, and decision-making interact.

Education

Monash University

Melbourne, Australia

Bachelor of Engineering (Honours) and Commerce

2021 – 2025

Majors: Electrical & Computer Systems Engineering; Mathematical Foundations of Econometrics

WAM: 74

Relevant Coursework: Time Series Econometrics; Probability and Statistical Inference; Econometric Theory; Linear Regression; Maximum Likelihood Estimation; Stochastic Models; Statistical Signal Processing; Signals and Systems; Control Systems; Optimisation; Real Analysis; Deep Learning.

Research Interests

Dynamic Systems: feedback systems, control theory, process dynamics, long-run system behaviour, sequential decision-making

Uncertainty & Statistical Modelling: time-series analysis, stochastic processes, econometrics, statistical inference, model validation, uncertainty-aware decision-making

Optimisation & Decision Systems: constrained optimisation, economic objective functions, model predictive control, target-setting, resource allocation, decision-support workflows

Applied Data Systems: high-dimensional industrial time-series data, multivariate monitoring, PCA, anomaly detection, interpretable diagnostics, production-grade analytics

Technical Skills

Programming: Python, MATLAB, SQL, C/C++, C#, .NET, Git/GitHub

Quantitative Methods: time-series analysis, regression, multivariate statistics, PCA, hypothesis testing, statistical diagnostics, stochastic processes, econometrics

Optimisation & Control: constrained optimisation, model predictive control, multivariable control, controller model validation, economic objective functions, steady-state optimisation

Engineering Systems: APC systems, DCS process data, Aspen DMC/DMC3, AVEVA PI, AVEVA ROMEo, Yokogawa DCS platforms

Current Development Focus: pandas, NumPy, scikit-learn, statsmodels, reproducible research notebooks, scientific computing, simulation, applied statistical learning

Selected Writing & Research

Final Year Research: Extending the Herd-Route System with Explorations in Ergodic Control Theory
2025

- Developed mathematical extensions to an ergodic control framework developed in the literature, focusing on convergence of long-run system behaviour, ergodic control concepts, optimisation, and stochastic-systems reasoning.
- Awarded Best Final Year Project Paper.

Professional Experience

Viva Energy Australia

Geelong, Australia

Graduate Advanced Process Control Engineer

Dec 2024 – Present

- Commissioned live refinery DMC controllers by analysing dynamic process responses, validating controller models, tuning constraints and objective functions, and monitoring closed-loop performance under real operating conditions.
- Diagnosed live APC applications across hydroprocessing and isomerisation units, analysing manipulated-variable / controlled-variable interactions, active constraints, disturbance propagation, controller degradation, and model-process mismatch.

- Analysed high-dimensional refinery time-series data using PCA, multivariate trend analysis, and process-variable relationship mapping to identify dominant operating modes and abnormal behaviour.
- Developed real-time APC support logic and refinery optimisation workflows across DCS, Aspen DMC/DMC3, AVEVA PI, and AVEVA ROMeo, linking process constraints, economic objectives, and controller target-setting.

Thornton Engineering

Engineering Student

Geelong, Australia

Jan 2021 – Mar 2023

- Rotated across site-floor operations, robotic welding, machine operations, project management, and procurement, developing practical exposure to manufacturing workflows, production constraints, and safety-critical industrial environments.
- Programmed and operated IGM welding robots using Lua, Python, and CAD-based design inputs, contributing to fabrication planning, machine operation, and layout improvements for large-scale engineering projects.

Writing & Communication

Economics Student Society of Australia

Writer / Editor

Melbourne, Australia

Jan 2024 – Jan 2025

- Wrote and edited economics-focused articles for online and print publication, translating macroeconomic and market-related topics for a broad student audience.
- Conducted interviews with public and private-sector participants and contributed to editorial planning, article allocation, and publication workflows.